Indian Institute of Information Technology, Allahabad



Digital Twin-Based Early Warning System for Cold Chain Disruption Detection

– Embedding Anomaly Detection Inside a Discrete-Event Simulation Loop (Topic 21)

Project Presentation

Presenter :

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Managing Corporate Entrepreneurship

Introduction

The Cold Chain Reality & High Stakes

- **Strict Thermal Bounds:** Cold-chain cargo (vaccines, biologics, perishable foods) demands strict, unbroken temperature control **(e.g., 2°C to 8°C).**
- **High Economic Impact:** A single temperature excursion compromises drug potency or causes food spoilage, generating billions in economic losses annually.
- **Public Health Risk:** Spoiled biologicals risk patient safety and cause massive pharmaceutical recalls.

The Traditional Failure Mode: Static Alarms

- **Status Quo:** Commercial supply chains rely on static threshold alarms (e.g., alert only if temperature exceeds 8°C for **>10–15 minutes**).
- **The Post-Hoc Flaw:** Alarms only sound after the thermal excursion has already occurred.
- **Irreversible Damage:** By the time an operator is notified, cumulative heat exposure has **already caused** irreversible biochemical degradation. Traditional monitoring is fundamentally reactive.

The Core Flaw: Waiting for a threshold breach is fundamentally reactive—by the time alarms sound, the damage is already irreversible!

Research Gap and Core Objectives

The Academic Gap & Proposed Solution

- **Prior literature:** focuses almost exclusively on post-hoc shelf-life calculation (Arrhenius degradation after the excursion).
- **Underexplored gap:** proactive, real-time anomaly detection embedded in an event-driven digital twin, with quantifiable early warning.
- **Proposed solution:** embed streaming detectors (Isolation Forest and LSTM-Autoencoder) in a discrete-event SimPy twin to catch precursors before irreversible failure.

RQ1: Early-Warning Runway

How many minutes of warning **($\Delta t_{lead} = t_{irreversibility} - t_{alarm}$)** can machine learning deliver before spoilage becomes irreversible?

RQ2: Classical vs. Deep Learning

How does a tree ensemble (Isolation Forest) compare with a deep sequence autoencoder (LSTM) on highly irregular sensor telemetry?

RQ3: Simulation Fidelity

Can an event-driven twin run these detectors in real time without silent divergence or cold-start lag versus offline batch evaluation?

Literature Review (2023–2026)

Paper and venue	Core contribution	Relevance
Vukšić et al. (2026) **Applied Sciences (MDPI)**	Hybrid physics-based digital twin + autoencoder for reefer containers	**~68 min** median lead time corroborates our **64 min** drift warning
Xie et al. (2025) **PLOS ONE**	Improved Isolation Forest for cold-chain fleet streams	Tree-ensemble streaming benchmark; context for point-wise F1
Meng et al. (2024) **J. Food Engineering**	Box-level digital twin temperature model with static alarms	The static-alarm baseline our system benchmarks against
Wu et al. (2023) **J. Industrial Information Integration**	Internet-of-Everything + digital-twin platform for cold-chain logistics	Precedent for embedding AI inference in twin pipelines
Defraeye et al. (2023) **Comput. Electron. Agric. / J. Food Eng.**	Review of food cold-chain digital twins and sensor dynamics	Gap: from reactive quality loss to live early warning

Vukšić et al. (2026): ~68 min lead time on reefers corroborates our 64-minute drift warning.

Data Strategy & Structural Proxy Framing

Empirical Telemetry Properties

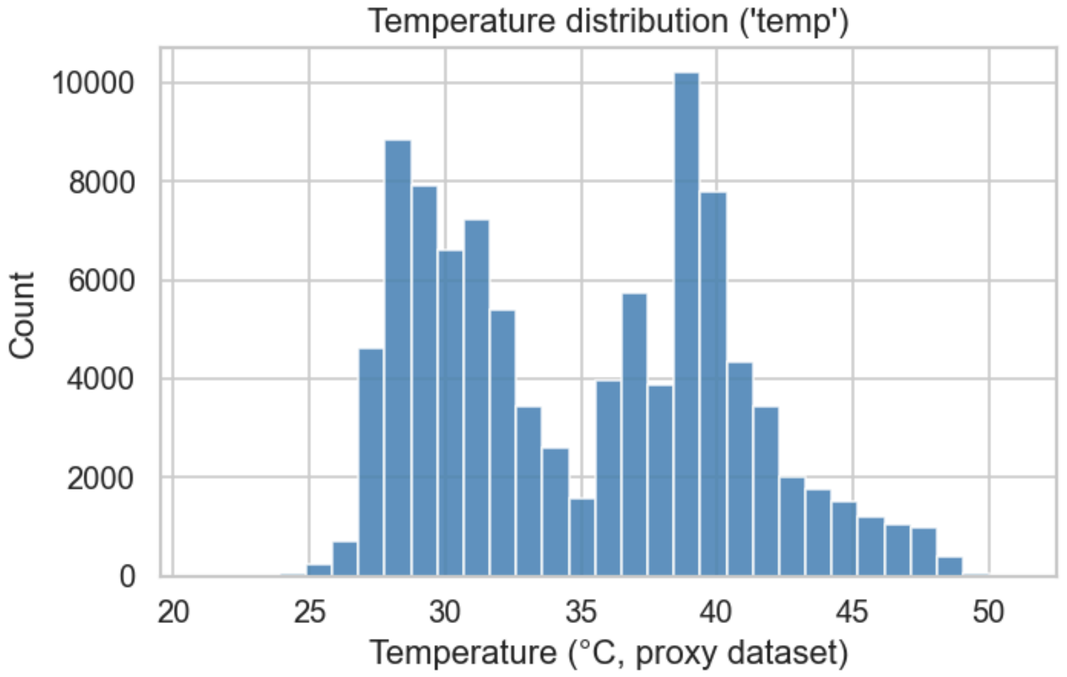
- **Kaggle IoT telemetry: 97,606** raw readings over **133** days (Jul 28 – Dec 8, 2018); 27,920 unique minute timestamps.

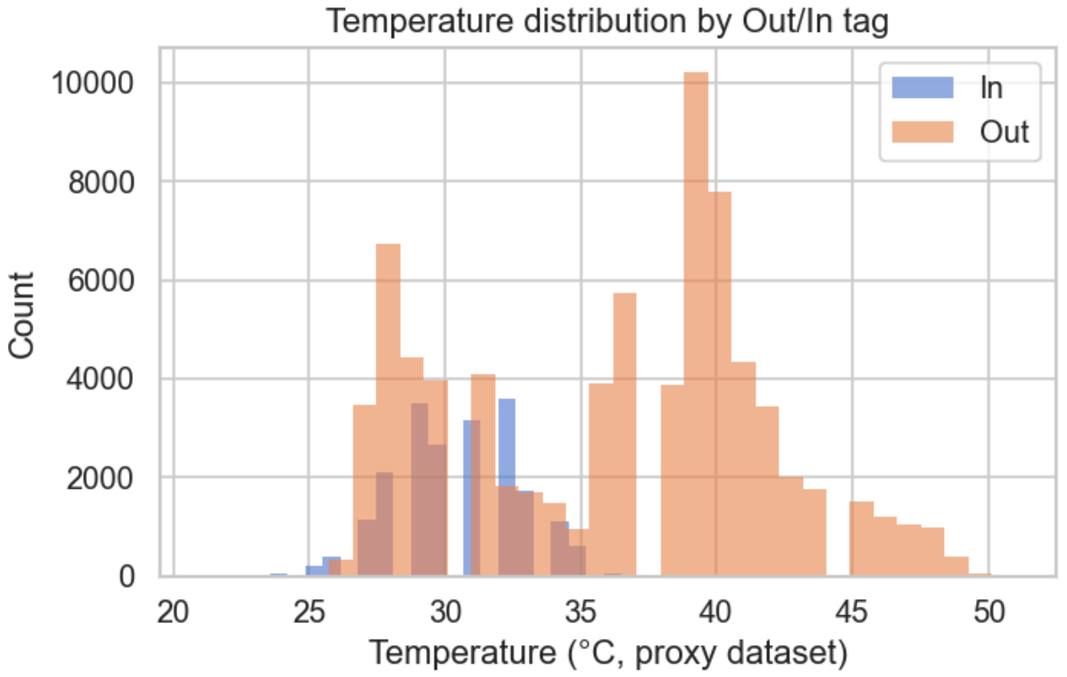
- **Extreme arrival irregularity:**
  - mean gap **117.8 s**
  - median **0.0 s**
  - CoV 46.23
  - dropouts up to 11.57 days.

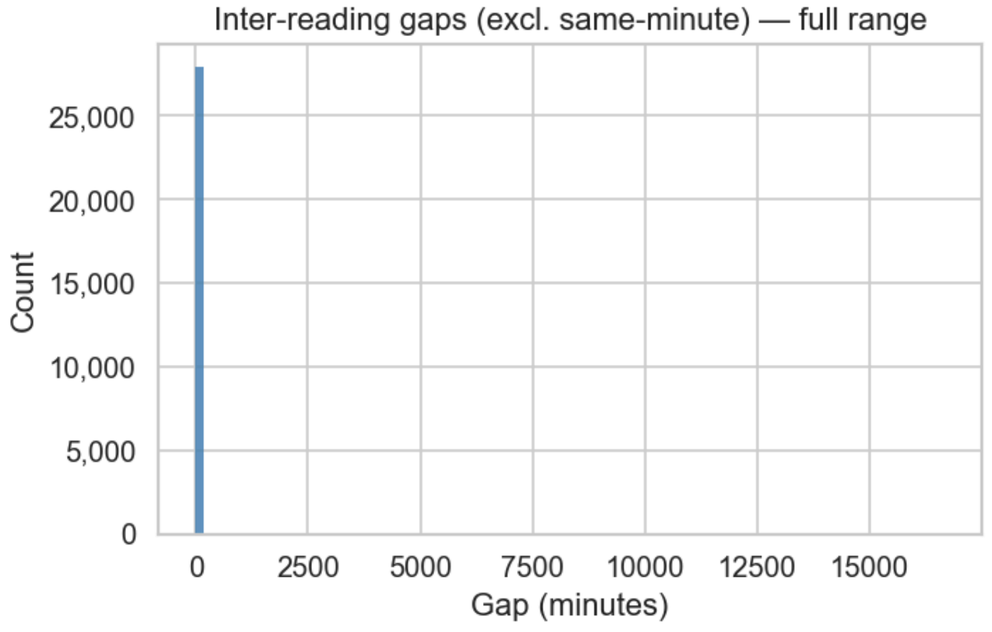
- **Dual streams:** ambient Out **(77,260 rows)** and internal enclosure proxy In **(20,345 rows)**.

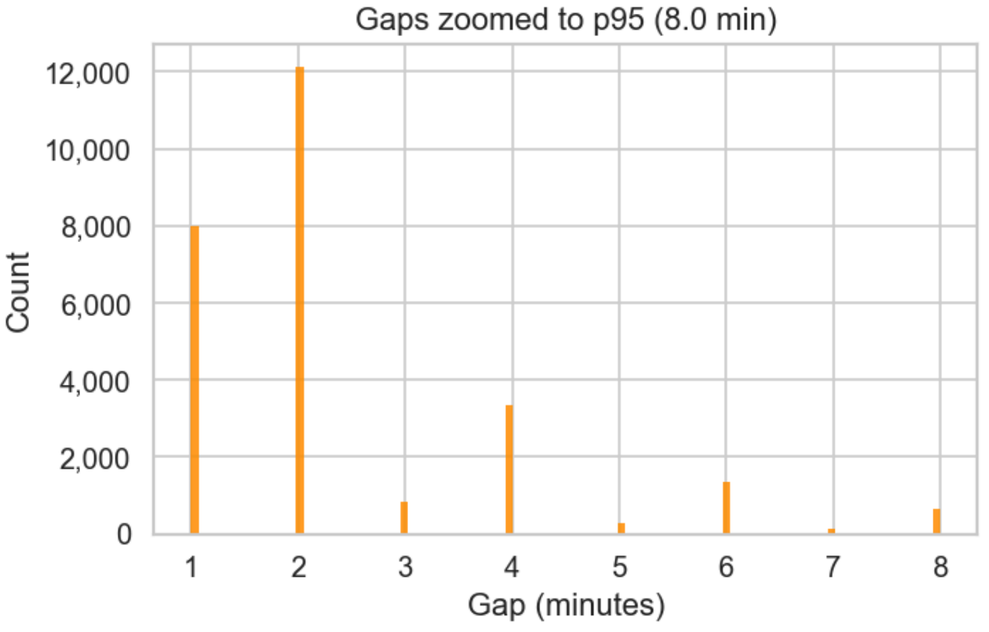
Academic integrity: structural proxy notice

Data comes from an office installation **("Room Admin")**, not live pharmaceutical reefers. It is used strictly as a structural proxy for irregular streaming pipelines, with no food-safety claims.


Temperature distribution ('temp')
Count
Temperature (°C, proxy dataset)


Temperature distribution by Out/In tag
In
Out
Count
Temperature (°C, proxy dataset)

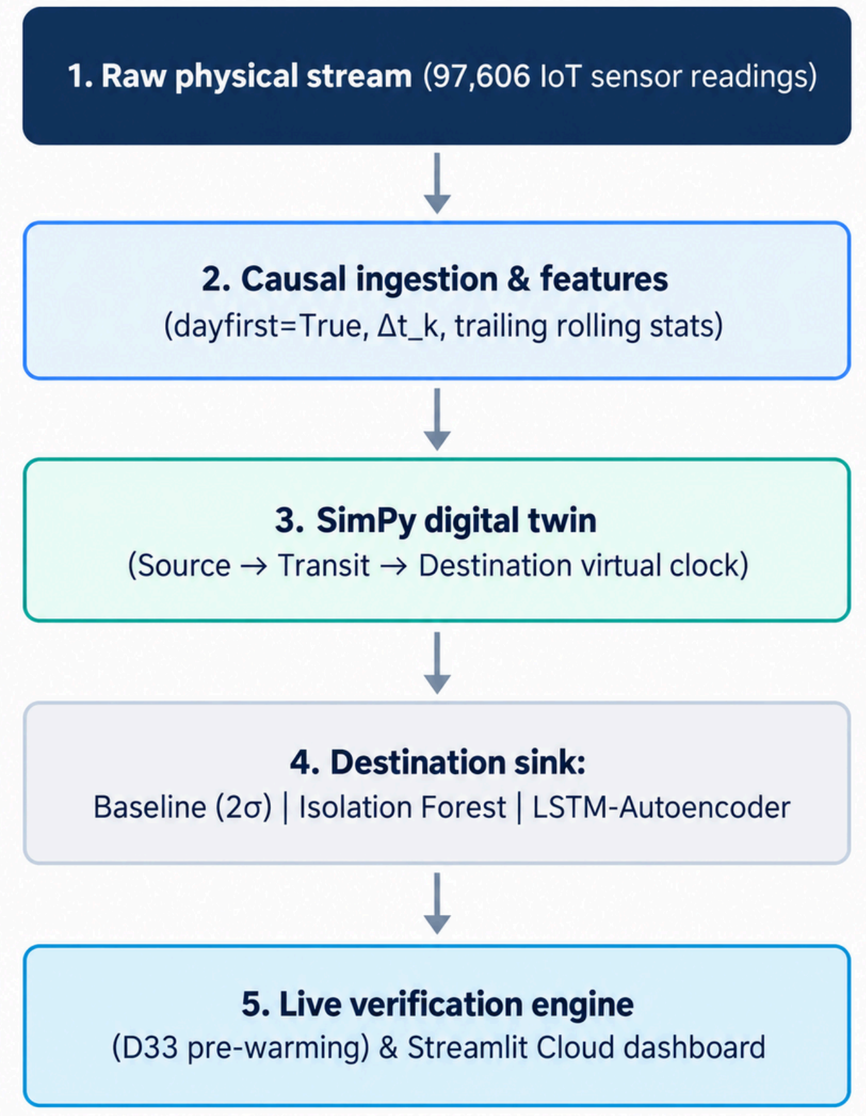

Inter-reading gaps (excl. same-minute) — full range
Count
Gap (minutes)


Gaps zoomed to p95 (8.0 min)
Count
Gap (minutes)

System Architecture & the Digital Twin

Discrete-Event Simulation Architecture

- **Causal ingestion:** day-first parsing (prevents 48.8% silent parse failures); trailing rolling stats with zero look-ahead leakage (D22).
- **SimPy 3-stage transit:** Source → Transit → Destination, driven by real inter-arrival gaps.
- **Embedded monitors:** Baseline (static 2σ: **47.7°C Out, 34.9°C In**), Isolation Forest (partition score), LSTM (reconstruction MSE).
- **Silent buffer pre-warming (D33):** windows are primed with preceding history, eliminating cold-start false alarms.



Experimental Setup & Anomaly Protocol

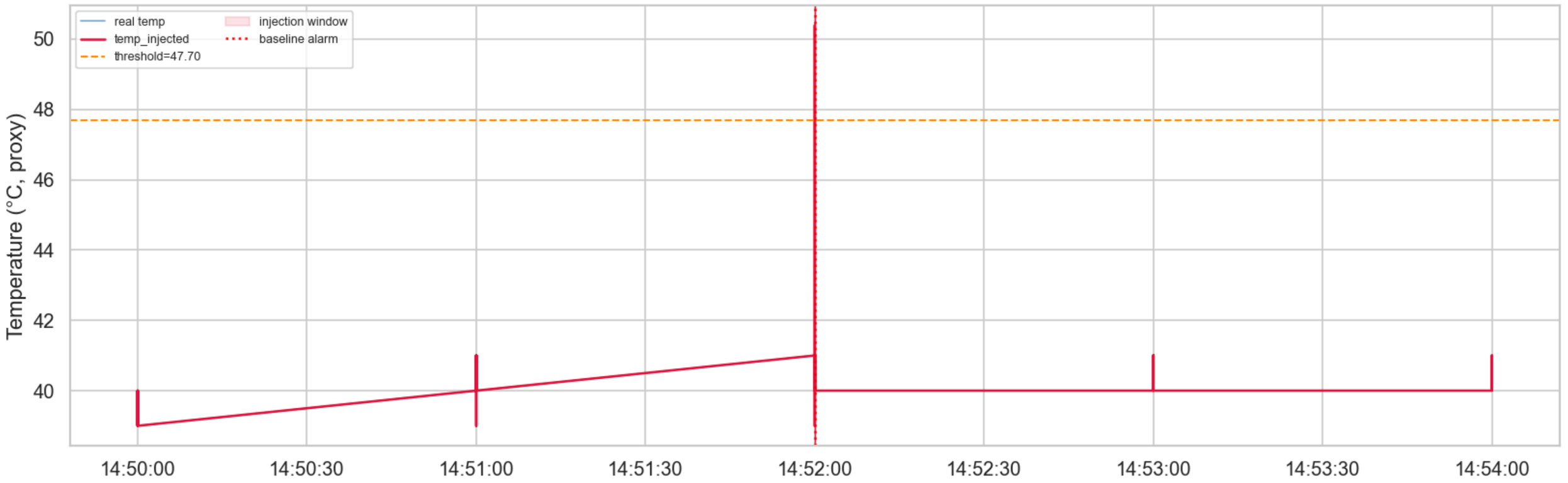
Controlled perturbation protocol (seed 42)

- **30 synthetic injections** (15 Out / 15 In), **10** per failure mode:
- **Step spikes:** abrupt **+5 to +15°C** jumps (door left open).
- **Linear drifts:** ramps of **+0.05 to +0.25°C/min** (compressor leak).
- **Sensor flatlines:** zero-variance persistence (frozen sensor).
- **Chronological split: 70% train / 30% test** (cutoff 2018-10-29 11:10:48).
- **Out-of-sample discipline: 24** injections fall in train (development only); all headline claims use the **6** untouched test injections.

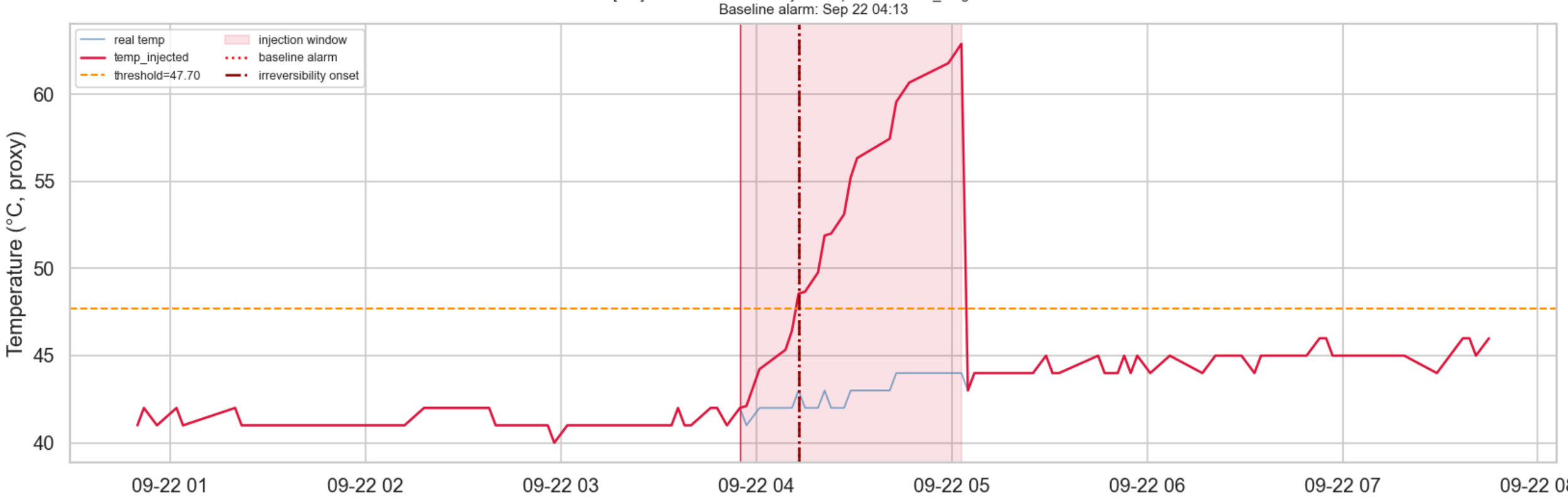
Lead time metric

$\Delta t_{lead} = t_{irreversability} - t_{alarm}$, where t_{irr} is a sustained 10-minute breach.
Positive = early warning; negative = late alarm.

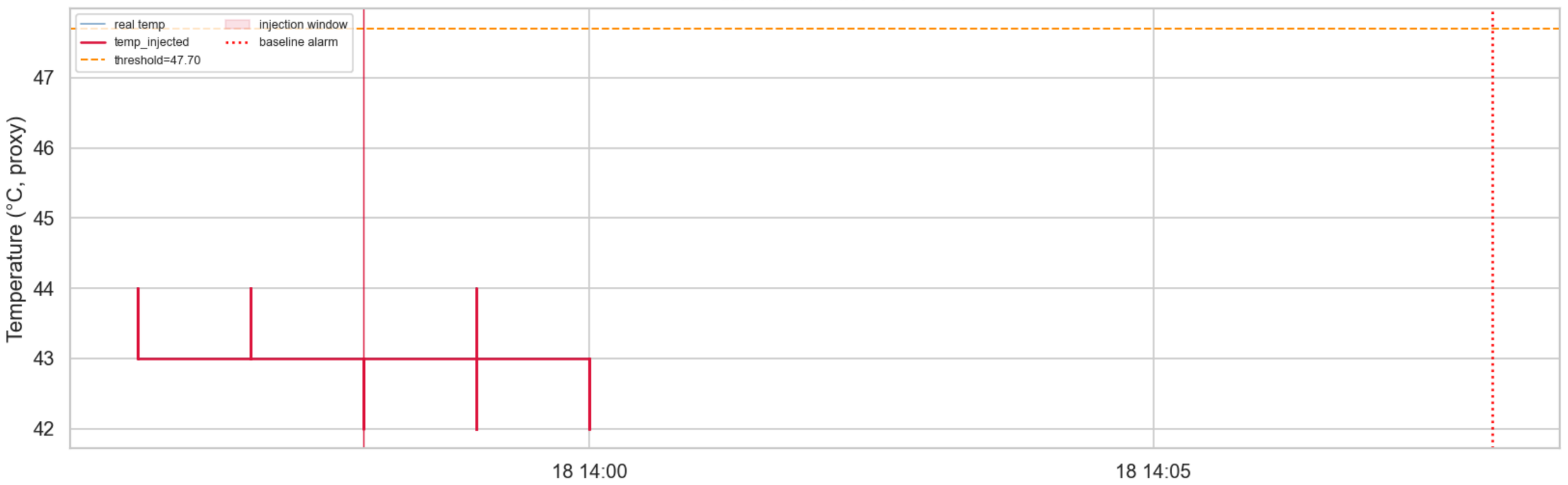
[Out] SYNTHETIC STEP injection | dur=7 mag=9.39
Baseline alarm: Oct 18 14:52



[Out] SYNTHETIC DRIFT injection | dur=18 final_mag=18.90
Baseline alarm: Sep 22 04:13



[Out] SYNTHETIC FLATLINE injection | dur=9 frozen_at=43.0
Baseline alarm: Oct 18 14:08



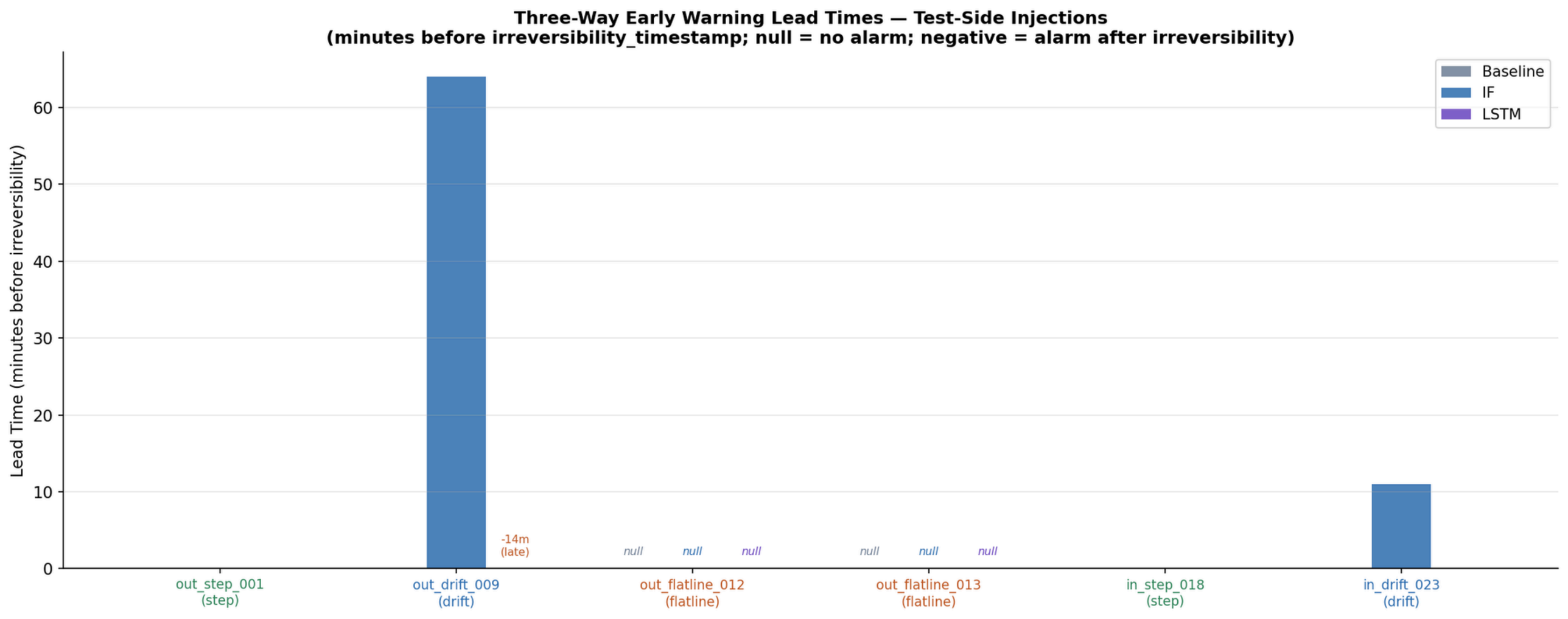
Core Result 1: Early-Warning Lead Times

Injection	Baseline	Isolation Forest	LSTM
out_step_001 (step)	0 min	0 min	0 min
out_drift_009 (drift)	0 min	+64 min	−14 min (late)
out_flatline_012	Missed	Detected	Detected
out_flatline_013	Missed	Detected	Missed
in_step_018 (step)	0 min	0 min	0 min
in_drift_023 (drift)	0 min	+11 min	0 min

Test-set injections only.
Isolation Forest wins on drift, warning 64 min (Out) and 11 min (In) before irreversibility.

Takeaways

- **Drift:** variance shifts before absolute thresholds are breached.
- **Step:** all models trigger together at onset (0 min).
- **Flatline:** static thresholds miss both; LSTM was 14 min late on Out drift.


Three-Way Early Warning Lead Times — Test-Side Injections
(minutes before irreversibility_timestamp; null = no alarm; negative = alarm after irreversibility)
Lead Time (minutes before irreversibility)
0
10
20
30
40
50
60
Baseline
IF
LSTM
-14m
(late)
null
null
null
null
null
null
out_step_001
(step)
out_drift_009
(drift)
out_flatline_012
(flatline)
out_flatline_013
(flatline)
in_step_018
(step)
in_drift_023
(drift)

Core Result 2: Trade-offs & Simulation Parity

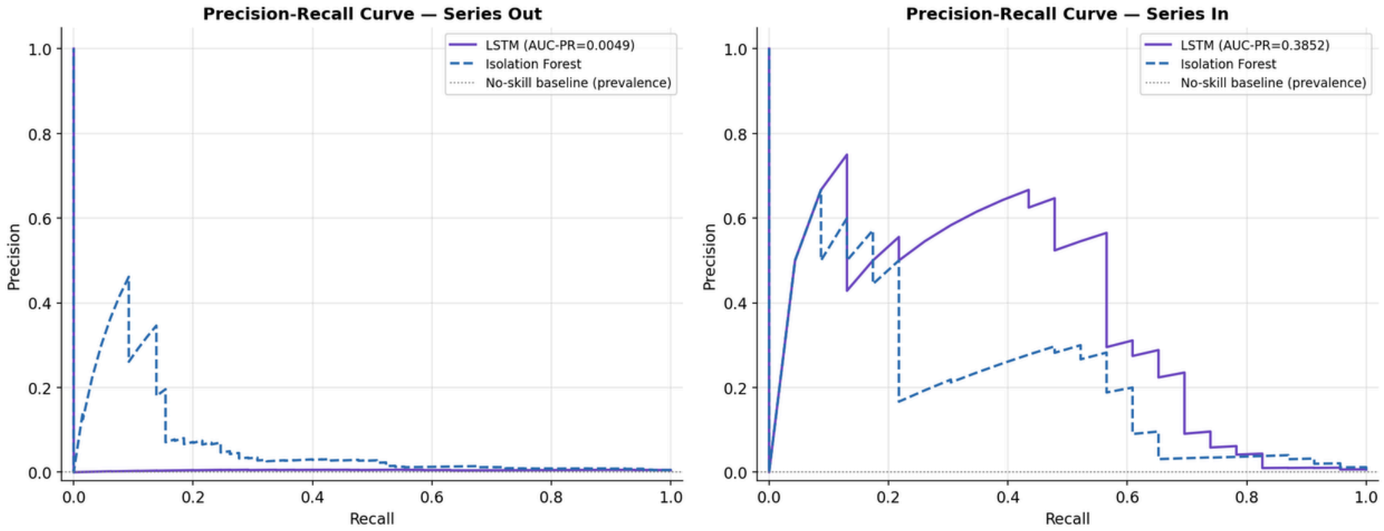
Isolation Forest: high-recall early tripwire

- Recall **80.00% (Out), 91.30% (In);**
  false positive rate **45.59%** Out, **26.98%** In.
- **Role**: aggressive frontline tripwire where missing an excursion is fatal.

LSTM-Autoencoder: precision-favoring verifier

- **In**: **13.11%** precision, **2.85%** FPR, AUC-PR **0.3852**.
  Out: recall **20.00%**, AUC-PR **0.0049.**
- **Role**: conservative secondary verifier that limits alarm fatigue.

Prevalence: anomalies are **~0.5%** of rows **(65/13,372 Out; 23/3,767 In)**, so low absolute F1 is inevitable. Lead time is the primary metric.



100% bit-for-bit live / batch parity

- 97,603 readings replayed in 6.03 s CPU (>1.9M× speedup), 0.00 s clock drift.
- 42 of 42 applicable test evaluations matched batch results.
- Live on Streamlit Cloud: Guided Tour and Free Explore modes.

Conclusion

Evidence-first risk management

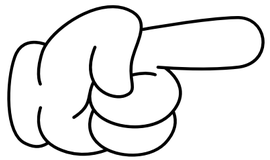
Honest reporting of trade-offs: we documented where deep learning lagged on drift and where classical ML excelled, avoiding fabricated 99% accuracy claims.

Disciplined architecture & auditability

Append-only decision log (35 entries, D1–D35). The audit caught 4 critical bugs (timestamp parse inversion, rolling leakage, cold starts) before they biased evaluation.

Technology adoption strategy

Classical ML (Isolation Forest) can outperform deep networks for edge streaming early warning (+64 min lead time) at low compute overhead.



Links

Github: https://github.com/Krishna200608/coldchain-ews-twin.git

Live Dashboard : coldchain-ews-twin.streamlit.app

Tech Stack

1 Core Languages & Environments

Python 3.11

Jupyter Notebooks / IPython

Google Colab (Cloud GPU)

2 Machine Learning & Deep Learning

TensorFlow / Keras (LSTM-Autoencoder)

Scikit-Learn (Isolation Forest, StandardScaler, Metrics)

Joblib

3 Digital Twin & Simulation

SimPy

Source → Transit → Destination

SimPy (Discrete-Event Simulatiion)

Custom 3-Stage Pipeline Architecture (Source → Transit → Destination)

4 Data Engineering & Scientific Computing

Pandas

NumPy

SciPy

KaggleHub / Kaggle API

5 Web Application & Deployment

Streamlit

Altair

Streamlit Community Cloud

6 Visualization & Plotting

Matplotlib

Seaborn

7 Automated Testing & Code Quality

Unittest

Ruff

Python-dotenv

8 Version Control & Engineering Audit

Git

GitHub

Architectural Decision (AD) Log (D1–D35)

References

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3. Meng, X., Xie, R., Liao, J., Shen, X., & Yang, S. (2024). A Cost-Effective Over-Temperature Alarm System for Cold Chain Delivery. Journal of Food Engineering, 368, 111914.

4. Wu, W., Shen, L., Zhao, Z., Harish, A. R., Zhong, R. Y., & Huang, G. Q. (2023). Internet of Everything and Digital Twin Enabled Service Platform for Cold Chain Logistics. Journal of Industrial Information Integration, 33, 100443.

5. Defraeye, T., Shrivastava, C., Berry, T., Verboven, P., Onwude, D., Schudel, S., et al. (2023). Digital Twins of Food Cold Chains: Advances, Challenges, and Opportunities. Computers and Electronics in Agriculture, 214, 108252.

6. Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. 2008 Eighth IEEE International Conference on Data Mining (ICDM), 413–422.

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8. Anand Jha, A. (2018). Temperature Readings: IoT Devices Telemetry Dataset. Kaggle Datasets (Structural Proxy Dataset).

THANK YOU!